

Matrix renewal and nonlinear recurrence: audited proof modules

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0.1 Random matrices: radius and direction must evolve together

For a quadratic in d parameters, the linear update is $U_{t+1} = A_{t+1}U_t$, where $A = I - \eta H_B$. Writing $U = rs$, with $\|s\| = 1$, gives

$$r^+ = r\|As\|, \quad s^+ = As/\|As\|. \quad (1)$$

Thus a matrix changes both the amplitude and the direction of the next update. Its log gains need not be independent. The appropriate analogue of the scalar moment $\mathbb{E}|a|^q$ is the growth rate of the positive operator

$$(P_q f)(s) = \mathbb{E} \left[\|As\|^q f \left(\frac{As}{\|As\|} \right) \right], \quad k(q) = \text{spr}(P_q). \quad (2)$$

Here spr denotes spectral radius. When the original angular chain is ergodic, identifies the top product exponent, and the pressure is differentiable at zero, $\gamma = (\log k)'(0)$ is the linear-product Lyapunov exponent. In one dimension, $k(q) = \mathbb{E}|a|^q$. Classical matrix renewal theory uses a positive root to describe large values of a contractive affine recursion [Kesten, 1973, Guivarc'h and Le Page, 2016]. The following result uses a negative root to describe small amplitudes of a locally expanding process supplied with a return mechanism.

Theorem 1 (Matrix central law with finite return cycles). *Suppose the linear product is locally expanding, $\gamma > 0$, and $k(-\alpha) = 1$ for some $\alpha > 0$. Assume the negative-order spectral and Markov renewal conditions in Assumption 1, including a bounded strictly positive eigenfunction and nonarithmetic angular mixing. Start an exact linear excursion below radius r_0 and restart it from the same independent entry law on its first exit above r_0 . If the mean excursion length is finite and the entry law has a finite inverse α -moment, then its invariant law satisfies*

$$\lim_{r \downarrow 0} \frac{\pi(\|U\| \leq r, U/\|U\| \in D)}{(r/r_0)^\alpha} = C\chi(D), \quad (3)$$

for angular continuity sets D , provided the tilted excursion has a positive probability of never returning to the upper boundary. Here $C > 0$ and χ is a probability measure on directions. The matrices may be noncommuting.

What this establishes. Finite return cycles create stationarity; the negative moment root determines how often stationary excursions reach the center. The proof tilts the joint radius–direction process, applies a Markov renewal theorem, and subtracts occupation after upper exit. It does not replace matrix products by Rayleigh quotients. For a genuine nonlinear SGD chain, Proposition 3 gives a precise transfer criterion in terms of its entry law and local transition kernel. Those conditions are additional to positive local growth and are not established for our neural networks.

Which density can lose its central peak? With an additional renewal-density regularity assumption, (3) gives radial density $p_R(r) \asymp r^{\alpha-1}$. If the angular law also has a regular density, the ambient density scales as $p_U(rs) \asymp r^{\alpha-d}$. For a strictly convex log pressure with roots $-\alpha$ and 0, $k(-1) < 1$ means $\alpha > 1$, whereas $k(-d) < 1$ means $\alpha > d$. A fixed coordinate integrates over directions and has no such universal threshold: Proposition 4 gives a stationary nonlinear SGD example with both inequalities but centrally unimodal fixed-coordinate marginals. This distinction motivates direct fixed-direction measurements in the neural experiments.

A Matrix renewal at a locally expanding center

A.1 Definitions and explicit assumptions

Let $A_1, A_2, \dots$ be iid invertible matrices. Work on a compact communicating class $\mathcal{S}$ of signed unit directions; projective space is sufficient when no statement depends on sign. For a nonzero vector, write $U = r_0 e^Z S$. Define the Markov additive process

$$S_{n+1} = \frac{A_{n+1} S_n}{\|A_{n+1} S_n\|}, \quad Z_{n+1} = Z_n + Y_{n+1}, \quad Y_{n+1} = \log \|A_{n+1} S_n\|. \quad (4)$$

Its joint direction-increment kernel is $K(s, ds', dy)$. The term ‘‘Markov additive’’ means precisely that this kernel depends on the current direction but not on the current log radius. Let P_q be (2), acting on continuous functions with the supremum norm, and let $k(q)$ be its spectral radius. In particular $P_0 1 = 1$ and $k(0) = 1$.

Assumption 1 (Negative-order renewal hypotheses). *The following conditions hold.*

1. For some $\alpha > 0$ there exist a continuous h , constants $0 < h_{\min} \leq h \leq h_{\max} < \infty$, and a probability measure χ such that

$$P_{-\alpha} h = h, \quad \chi P_{-\alpha} = \chi, \quad \chi(h) = 1, \quad k(-\alpha) = 1. \quad (5)$$

For some $t \in (0, \alpha)$, $P_{-\alpha+t}$ is bounded on $C(\mathcal{S})$ and $k(-\alpha + t) < 1$.

2. The probability kernel

$$\hat{K}(s, ds', dy) = e^{-\alpha y} \frac{h(s')}{h(s)} K(s, ds', dy) \quad (6)$$

has a positive Harris recurrent angular chain, with invariant law $\hat{\chi}(ds) = h(s)\chi(ds)$, and each one-step angular law $\hat{P}(s, \cdot)$ is absolutely continuous with respect to $\hat{\chi}$. Under $\hat{\chi}$,

$$\hat{\mathbb{E}}|Y_1| < \infty, \quad \hat{\mathbb{E}}Y_1 = -m < 0. \quad (7)$$

3. The tilted Markov additive process is nonarithmetic: there is no $b > 0$ and measurable $c : \mathcal{S} \rightarrow [0, b)$ such that $Y_1 \in c(S_0) - c(S_1) + b\mathbb{Z}$ almost surely under its stationary angular law.

These are hypotheses on the actual random matrices, not conclusions from a positive one-step scalar log gain. Absolute continuity in the second condition is a convenient sufficient condition extending the almost-every-start Markov renewal theorem to every starting direction. Singular or finitely supported matrix laws may require a different all-start renewal theorem; the present statement does not silently cover them. If the dominant eigenvalue and eigenfunction are differentiable at $-\alpha$, differentiating $P_q h_q = k(q)h_q$ and integrating against χ gives $k'(-\alpha) = \chi(P'_{-\alpha} h) = -m$.

Thus $m = -(\log k)'(-\alpha)$ in that case. Strict convexity of $\log k$ between the two roots guarantees $k(-\alpha + t) < 1$.

Let $\tau = \inf\{n \geq 0 : Z_n \geq 0\}$ and let ζ be an entry law on $(-\infty, 0) \times \mathcal{S}$. We impose

$$L := \mathbb{E}_\zeta \tau \in (0, \infty), \quad \mathbb{E}_\zeta e^{-\alpha Z_0} < \infty. \quad (8)$$

At the transition that would exit, the regenerative reference chain instead draws its next state from an independent copy of ζ . Consequently each cycle contains states numbered $0, \dots, \tau - 1$. This construction specifies a reference process. Probabilistic regeneration of actual SGD is addressed separately below.

A.2 A uniform occupation bound derived from the spectrum

Lemma 2 (Tilted interval potentials are uniformly bounded). *Under Assumption 1(1), for each $b > 0$ there is $C_b < \infty$ such that*

$$\sup_{z,s,x} \widehat{\mathbb{E}}_{z,s} \sum_{n \geq 0} \mathbf{1}_{\{Z_n \in (x, x+b]\}} \leq C_b. \quad (9)$$

Proof. The exact change of measure gives

$$\widehat{\mathbb{E}}_s e^{t(Z_n - Z_0)} = \frac{P_{-\alpha+t}^n h(s)}{h(s)} \leq C \rho^n \quad (10)$$

for some $\rho < 1$ and $C < \infty$. Indeed, choose ρ strictly between $k(-\alpha + t)$ and 1; the spectral-radius formula bounds the operator powers by a constant times ρ^n , and h is bounded above and away from zero. If $Z_0 \in (x, x + b]$, exponential Markov inequality yields

$$\widehat{\mathbb{P}}_{Z_0,s}(Z_n \in (x, x + b]) \leq e^{tb} C \rho^n.$$

Summing over n bounds occupation by $e^{tb} C / (1 - \rho)$. For any other initial state, apply this bound at its first visit to the interval, using the strong Markov property. If there is no first visit the occupation is zero. This proves the claim, uniformly over both direction and log radius. $\square$

A.3 Unrestricted and killed renewal asymptotics

Write $U_{z,s}$ and $\widehat{U}_{z,s}$ for the unrestricted occupation measures of (4) under its original and tilted laws. For n steps, multiplying (6) along a path gives the exact likelihood ratio

$$\left. \frac{d\widehat{\mathbb{P}}_{z,s}}{d\mathbb{P}_{z,s}} \right|_{\mathcal{F}_n} = e^{-\alpha(Z_n - z)} \frac{h(S_n)}{h(s)}. \quad (11)$$

Consequently

$$U_{z,s}(dx, ds') = e^{\alpha(x-z)} \frac{h(s)}{h(s')} \widehat{U}_{z,s}(dx, ds'). \quad (12)$$

Apply the Markov renewal theorem of Alsmeyer [1997, Theorem 1] to the positive-drift process $-Z$. For an angular set D and a finite log interval, its conclusion, together with the exponential test weight in (12), gives

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} U_{z,s}((x, x + b] \times D) = e^{-\alpha z} h(s) \frac{e^{\alpha b} - 1}{\alpha m} \chi(D). \quad (13)$$

Initially the theorem gives the limit for $\widehat{\chi}$ -almost every starting direction. To obtain it for an arbitrary s , split the tilted occupation into its time-zero atom and its potential after the first step. The time-zero atom eventually lies outside the translated interval. The next direction has a law absolutely continuous with respect to $\widehat{\chi}$, so the almost-everywhere limit applies after this step. Lemma 2, together with the bounded test weight $e^{\alpha u}/h(s')$, permits dominated convergence over this first step. This proves the limit for every s and every finite initial z . The angular factor is χ , since division by $h(s')$ cancels the $h(s')$ in $\widehat{\chi}$.

Moreover, (12) and the same lemma give a constant C'_b such that, for all x, z, s ,

$$e^{-\alpha x} U_{z,s}((x, x+b] \times \mathcal{S}) \leq C'_b e^{-\alpha z} h(s). \quad (14)$$

This bound is the domination needed for integration over entry and exit states. It has been proved, rather than included as an unverified “uniform renewal” assumption.

Proof of Theorem 1. Let G_ζ denote expected occupation before τ . The strong Markov property gives, on bounded log intervals,

$$G_\zeta = U_\zeta - \mathbb{E}_\zeta U_{Z_\tau, S_\tau}. \quad (15)$$

Finite mean exit time implies $\tau < \infty$ almost surely under $\mathbb{P}_\zeta$. The entry moment in (8), bounded h , and $Z_\tau \geq 0$ permit integrating (13) in both terms by (14). Hence

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} G_\zeta((x, x+b] \times D) = D_\alpha \frac{e^{\alpha b} - 1}{\alpha m} \chi(D), \quad (16)$$

where

$$D_\alpha = \mathbb{E}_\zeta [e^{-\alpha Z_0} h(S_0) - e^{-\alpha Z_\tau} h(S_\tau)]. \quad (17)$$

Its sign cannot be read by comparing endpoint radii alone, because directions change. For each fixed start, summing the stopped likelihood ratio (11) over events $\{\tau = n\}$ instead gives

$$\mathbb{E}_{z,s} [e^{-\alpha Z_\tau} h(S_\tau); \tau < \infty] = e^{-\alpha z} h(s) \widehat{\mathbb{P}}_{z,s}(\tau < \infty).$$

Therefore

$$D_\alpha = \mathbb{E}_\zeta [e^{-\alpha Z_0} h(S_0) \widehat{\mathbb{P}}_{Z_0, S_0}(\tau = \infty)]. \quad (18)$$

The stated positive-survival condition makes $D_\alpha > 0$. It is enough for a positive entry mass to be sufficiently far below the upper boundary: by (10),

$$\widehat{\mathbb{P}}_{z,s}(\tau < \infty) \leq \sum_{n \geq 1} \widehat{\mathbb{P}}_{z,s}(Z_n \geq 0) \leq \frac{C\rho}{1-\rho} e^{tz},$$

which is strictly below one for all sufficiently negative z .

The regenerative occupation formula

$$\pi(F) = \frac{1}{L} \mathbb{E}_\zeta \sum_{n=0}^{\tau-1} \mathbf{1}_{\{(Z_n, S_n) \in F\}} \quad (19)$$

defines an invariant probability. For a bounded test function, summing its expected one-step change over a cycle telescopes to the difference between two cycle starts, both with law ζ ; hence its integral against $\pi P - \pi$ is zero. To pass from intervals to the half-line $(-\infty, x]$, partition it into intervals $(x - (j+1)b, x - jb]$. The bound (14) dominates their rescaled occupations by a summable geometric sequence in j . Summing (16) therefore yields

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} \pi(Z \leq x, S \in D) = \frac{D_\alpha}{\alpha m L} \chi(D). \quad (20)$$

Substituting $x = \log(r/r_0)$ proves the theorem, with $C = D_\alpha/(\alpha m L)$. $\square$

The theorem proves a small-ball asymptotic. Differentiating an asymptotic cumulative distribution without further hypotheses is invalid. A radial-density conclusion requires the corresponding renewal-density result, for example a suitable smoothing condition giving convergence of local densities. Angular density conclusions additionally require regularity of χ .

A.4 A noncommuting law that verifies every renewal hypothesis

The preceding assumptions are realizable even by symmetric random matrices. For $d \geq 2$, take

$$A = M(I - 2vv^\top), \quad v \sim \text{Unif}(S^{d-1}), \quad M \sim \frac{1}{4} \text{Unif} [.49, .51] + \frac{3}{4} \text{Unif} [1.99, 2.01], \quad (21)$$

with independent draws at each update. Independent reflections generically do not commute. Their norm gain is M , so $P_q = m(q)R$, where $m(q) = \mathbb{E}M^q$ and R is the angular reflection kernel. Thus $h = 1$, $k(q) = m(q)$, and χ is uniform surface probability. This scalar formula is special to this example; the matrix theorem itself retains the coupled angular dynamics.

For completeness the reflection kernel has, with respect to uniform surface probability, density

$$R(s, ds') = \|s - s'\|^{2-d} \chi(ds') \quad (s' \neq s). \quad (22)$$

To verify it, put $s = e_1$ and let θ be the angle between s and s' . Reflection gives $|v^\top s| = \sin(\theta/2)$. Changing variables from the spherical coordinate of v gives the angle density $c_d \cos^{d-2}(\theta/2)$, whereas a uniform s' has angle density $c_d \sin^{d-2} \theta$. Their ratio is $(2 \sin(\theta/2))^{2-d} = \|s - s'\|^{2-d}$. In particular $R(s, \cdot) \geq 2^{2-d} \chi(\cdot)$, so it is a uniformly mixing Harris kernel, and its symmetry verifies invariance of χ . Its one-step laws are absolutely continuous.

The top product norm equals $\prod_{j=1}^n M_j$, since every remaining factor is orthogonal. Hence

$$\gamma = \mathbb{E} \log M \geq \frac{1}{4} \log(.49) + \frac{3}{4} \log(1.99) > 0, \quad m(-1) \leq \frac{1}{4(.49)} + \frac{3}{4(1.99)} < 1.$$

Strict log convexity and the positive mass below one imply a unique negative root $m(-\alpha) = 1$ with $\alpha > 1$ and $m(-\alpha + t) < 1$ for every $t \in (0, \alpha)$. Tilting changes only the scalar multiplier law; it leaves R unchanged. The tilted log increment has bounded support, a density, and strictly negative mean $m'(-\alpha) < 0$, so all spectral and nonarithmetic hypotheses hold.

Choose the entry log radius $Z_0 = -1$ and an independent uniform direction. Since the original log increments are iid with mean $\gamma > 0$ and are bounded above by $\log 2.01$, Wald's identity at $\tau \wedge n$ gives

$$\gamma \mathbb{E}(\tau \wedge n) = \mathbb{E}Z_{\tau \wedge n} + 1 \leq \log 2.01 + 1.$$

The strong law gives almost-sure exit; monotone convergence now gives finite mean exit time. The entry inverse moment is e^α . Under the tilted law $e^{\alpha Z_n}$ is a nonnegative martingale; optional stopping yields $\widehat{\mathbb{P}}_{-1,s}(\tau < \infty) \leq e^{-\alpha} < 1$. The central-law coefficient is therefore strictly positive. All hypotheses of Theorem 1 have been verified for this noncommuting law.

The matrices are symmetric, so they can be written $A = I - \eta H$ for symmetric sample Hessians. In $d = 2$, $\mathbb{E}(I - 2vv^\top) = 0$, so $\mathbb{E}H = I/\eta$ is positive definite even though individual sample Hessians need not be. The independent restart remains an explicit return mechanism of the reference process; it is not silently identified with an SGD update.

A.5 An exact, conditional transfer statement for nonlinear SGD

Proposition 3 (Nonlinear occupation transfer). *Consider a nonlinear chain that has genuine regenerative cycles of finite mean length T , with cycle endpoints outside the local domain $\mathcal{D} =$*

$\{(z, s) : z < 0\}$. Let Q be its actual transition kernel killed on leaving $\mathcal{D}$, and let L_0 be the kernel of (4) killed at its upper exit. Let G be the expected measure of all local visits in one complete cycle, and ν the expected measure of every entry into $\mathcal{D}$ in that cycle. Put

$$\mathcal{J} = \nu + G(Q - L_0), \quad H(z, s) = e^{-\alpha z} h(s) \widehat{\mathbb{P}}_{z,s}(\tau = \infty). \quad (23)$$

Assume the killed linear chain exits from G -almost every starting point and that

$$\int e^{-\alpha z} |\mathcal{J}|(dz, ds) < \infty, \quad D_Q := \int H d\mathcal{J} > 0. \quad (24)$$

Under Assumption 1, the genuine invariant law has the same central exponent:

$$\lim_{r \downarrow 0} \frac{\pi(\|U\| \leq r, S \in D)}{(r/r_0)^\alpha} = \frac{D_Q}{\alpha m \mathbb{E}T} \chi(D). \quad (25)$$

Proof. Counting each visit as either an entry or the successor of a previous local visit yields $G = \nu + GQ$. Thus $G = \mathcal{J} + GL_0$ exactly, and

$$\sum_{j=0}^{n-1} \mathcal{J} L_0^j = G - GL_0^n.$$

Since G is finite and the killed linear chain exits G -almost everywhere, GL_0^n tends to zero in total variation. For each fixed bounded log interval, the killed potential is bounded by its unrestricted potential, and (14) shows absolute integrability against $|\mathcal{J}|$ by (24). We may therefore pass to the infinite sum and then integrate the killed asymptotic against the signed source. The coefficient is D_Q . Summing intervals as in the preceding proof and dividing occupation by the complete nonlinear cycle length proves (25). $\square$

The source is signed. A nonnegative entry law does not guarantee $D_Q > 0$, because nonlinear correction can cancel its leading term. The source condition can be checked without assuming the desired central exponent. For example, suppose $V_p(z, s)$ is comparable to e^{-pz} for some $0 < p < \alpha$ and satisfies

$$QV_p \leq \rho_p V_p, \quad \rho_p < 1, \quad \nu(e^{-\alpha z}) < \infty, \quad (26)$$

and

$$\int e^{-\alpha z'} |Q - L_0|((z, s), d(z', s')) \leq CV_p(z, s). \quad (27)$$

Decomposing the cycle into its actual local excursions gives $G = \nu \sum_{n \geq 0} Q^n$ and hence $G(V_p) \leq \nu(V_p)/(1 - \rho_p) < \infty$. Equations (26)–(27) then imply (24)'s integrability assertion. An eigenfunction h_p with $P_{-p}h_p = k(-p)h_p$ and $k(-p) < 1$ suggests $V_p = e^{-pz}h_p(s)$ for this independent drift calculation.

A Taylor expansion $F_B(u) = A_B u + O(\|u\|^{1+\delta})$ does not by itself prove (27): nearby point masses have order-one total variation distance. Smooth transition laws or a different perturbative renewal theorem are needed. Fixed nondegenerate additive noise also destroys the multiplicative approximation as $r \downarrow 0$. For that case an intermediate small-noise annulus, with estimates uniform in the noise scale, must replace the literal zero-radius limit. We do not claim these transfer conditions for the neural experiments.

B An exact SGD construction and a projection counterexample

This example verifies stationarity and positive local matrix-product growth in every dimension. It also rules out a universal inference from matrix negative moments to fixed-coordinate bimodality.

Proposition 4 (Stationarity past the local matrix thresholds). *For every $d \geq 2$ and $\rho, \sigma > 0$, there is an explicit online nonquadratic regression loss for which fixed-step SGD has a unique attracting invariant probability with finite second moment, while its linearization at the population minimizer satisfies*

$$\gamma \geq \frac{\log 2}{2d} > 0, \quad k(-1) \leq 1 - \frac{1}{8d} < 1, \quad k(-d) = \frac{7}{8} < 1. \quad (28)$$

The pressure has a unique nonzero root $k(-\alpha) = 1$ with $\alpha > d$. Nevertheless, for $d \geq 3$ every fixed one-dimensional projection of the invariant law has a centrally unimodal density.

Proof. At each update independently sample

$$v \sim \text{Unif}(S^{d-1}), \quad h = \begin{cases} 1/2 & \text{with probability } 1/4, \\ 3 & \text{with probability } 3/4, \end{cases} \quad e \sim N(0, \sigma^2).$$

Let $a = \sqrt{h}v$, $q = (a^\top u)^2/2$, and use step size one on

$$\ell(u; a, e) = \frac{1}{2}(a^\top u - e)^2 - \frac{1}{2}\{q - \rho \log(1 + q/\rho)\}. \quad (29)$$

The centered residual makes the population loss, apart from a constant, the expectation of $q/2 + (\rho/2)\log(1 + q/\rho)$. This expression is positive when $q > 0$, and the random directions span the space, so the unique population minimizer is zero. Differentiation gives the exact SGD map

$$u^+ = \left(I - h \left[1 - \frac{q}{2(\rho + q)} \right] vv^\top \right) u + e\sqrt{h}v. \quad (30)$$

Local matrix growth. The derivative at zero is $A = I - hvv^\top$. For any fixed unit s , set $T = (v^\top s)^2$, so $\mathbb{E}T = 1/d$. The norm gain is

$$G^2 = \|As\|^2 = 1 + h(h - 2)T, \quad \frac{1}{2} \leq G \leq 2.$$

Isotropy makes this conditional gain law independent of the entire past, including the evolving direction of a matrix product. Thus its log gains are iid. The chord inequalities

$$\frac{1}{2} \log(1 + 3T) \geq T \log 2, \quad \frac{1}{2} \log(1 - 3T/4) \geq -T \log 2$$

give $\mathbb{E} \log G \geq (\log 2)/(2d) > 0$. For each deterministic nonzero initial vector the strong law identifies the product exponent with this expectation. Applying this simultaneously to a finite orthonormal basis, and bounding the operator norm above by the Frobenius norm and below by each column norm, proves that the top matrix-product exponent has the same value.

Negative pressure. For every real z , isotropy gives $P_z 1 = m(z)1$, where $m(z) = \mathbb{E}G^z$. Positivity yields $\|P_z\|_\infty = m(z)$, so $k(z) = m(z)$ exactly. The convex function

$$F(t) = \frac{1}{4}(1 - 3t/4)^{-1/2} + \frac{3}{4}(1 + 3t)^{-1/2}$$

has $F(0) = 1$ and $F(1) = 7/8$, hence $F(t) \leq 1 - t/8$ for $0 \leq t \leq 1$. Consequently $k(-1) = \mathbb{E}F(T) \leq 1 - 1/(8d)$. For any invertible matrix B , polar integration of $e^{-\|Bx\|^2}$ gives

$$\mathbb{E}_{s \sim \text{Unif}(S^{d-1})} \|Bs\|^{-d} = |\det B|^{-1}. \quad (31)$$

Since $|\det(I - hvv^\top)| = |1 - h|$, isotropy and (31) imply $k(-d) = \frac{1}{4} \cdot 2 + \frac{3}{4} \cdot \frac{1}{2} = 7/8$. All moments exist because $G \in [1/2, 2]$. The function $\log m$ is strictly convex because $\log G$ is nonconstant; $m(0) = 1$ and $m'(0) = \gamma > 0$. Since $G < 1$ with positive probability, $m(-s) \rightarrow \infty$ as $s \rightarrow \infty$. There is therefore a unique nonzero negative root, and $m(-d) < 1$ puts it at $-\alpha < -d$.

Global nonlinear return. Rewrite (30) exactly as

$$u^+ = A_\infty u + b(u; h, v) + e\sqrt{h}v, \quad A_\infty = I - \frac{h}{2}vv^\top, \quad (32)$$

$$b(u; h, v) = -\frac{\rho h}{2(\rho + q)}vv^\top u, \quad \|b(u; h, v)\| \leq B_0 := \sqrt{3\rho/8}. \quad (33)$$

The bound follows by maximizing $x/(\rho + hx^2/2)$ for $x \geq 0$. With $c = h/2$, $\mathbb{E}[c(2 - c)] = 43/64$, and hence

$$\mathbb{E}\|A_\infty u\|^2 = \theta\|u\|^2, \quad \theta = 1 - \frac{43}{64d} < 1.$$

The centered independent residual removes cross terms and contributes $\mathbb{E}\|e\sqrt{h}v\|^2 = 19\sigma^2/8$. Young's inequality gives

$$\mathbb{E}[\|u^+\|^2 \mid u] \leq (1 + \epsilon)\theta\|u\|^2 + (1 + \epsilon^{-1})B_0^2 + 19\sigma^2/8.$$

Choosing $\epsilon = (1 - \theta)/(2\theta)$ makes the first coefficient $\lambda = (1 + \theta)/2 < 1$. This is a geometric drift bound for $V(u) = 1 + \|u\|^2$. The map is Feller. Iterating the bound gives uniform second moments for the time marginals from any fixed initial state; their Cesàro averages are tight, and any weak limit is invariant. Lower semicontinuity gives a finite invariant second moment.

Irreducibility and uniqueness. Conditional on h , the one-step kernel has, for $y \neq u$, density

$$p_h(u, y) = \frac{2\varphi_{\sigma\sqrt{h}}(r - \mu_h(w, u))}{|S^{d-1}|r^{d-1}}, \quad r = \|y - u\|, \quad w = (y - u)/r, \quad (34)$$

where φ_s is the centered Gaussian density of standard deviation s and

$$\mu_h(w, u) = -h \left[1 - \frac{h(w^\top u)^2/2}{2(\rho + h(w^\top u)^2/2)} \right] w^\top u.$$

To derive (34), condition on the random axis and write the signed displacement along it as a Gaussian with mean $\mu_h(v, u)$ and variance $\sigma^2 h$. The two signed axes $w, -w$ give equal contributions, and polar coordinates give the displayed Jacobian. The density is strictly positive. On compact source and target sets, r and the absolute Gaussian argument are uniformly bounded, so it has a positive uniform lower bound off the diagonal; the diagonal is Lebesgue-null. Every compact set is therefore small, the chain is Lebesgue-irreducible, and minorizing a ball on itself gives aperiodicity. The drift and small-set condition imply unique positive Harris recurrence and convergence from every starting point [Hairer and Mattingly, 2011].

Fixed projections remain unimodal. The update law commutes with every orthogonal transformation. Uniqueness therefore makes its invariant distribution rotationally invariant. We may write $U = RS$, where S is uniform on the sphere independently of $R > 0$. For any unit direction v_0 and $d \geq 3$, the conditional density of $v_0^\top U$ is

$$f(x \mid R) = \frac{c_d}{R} \left(1 - \frac{x^2}{R^2}\right)^{(d-3)/2} \mathbf{1}_{\{|x| < R\}}. \quad (35)$$

It is symmetric and nonincreasing in $|x|$: it is uniform on $(-R, R)$ when $d = 3$. Mixing over R preserves this monotonicity. Thus every fixed projection is centrally unimodal despite (28) and $\alpha > d$. $\square$

This proposition establishes post-*local*-Lyapunov stationarity without claiming the central renewal law for the noisy chain. In particular, fixed additive noise cannot be neglected in the limit $\|u\| \downarrow 0$. The sample loss in this particular example is nonquadratic but convex along its observed coordinate: writing $x = a^\top u$ and $t = x^2/(2\rho)$, its second derivative is $1/2 + (1-t)/(2(1+t)^2) \geq 7/16$. It should not be used as the separate nonconvex example.

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